COMPUTER VISION & OBJECT DETECTION

Smart Waste Detection

Real-time multi-class material categorization using YOLO26 Nano.

Local Project (Pre-Release)

THE PROBLEM

Inefficient Sorting

- **Contamination:** Mixed waste streams ruin recyclable batches.
- **Manual Bottlenecks:** Human sorting is slow and hazardous.
- **Ecological Impact:** Missed recovery of biodegradable matter.



Solution Objectives

Instant Vision

Real-time automated inference.

6-Class Categorization Framework

Biodegradable | Cardboard | Glass | Metal | Paper | Plastic

Application Stack



Python



FastAPI

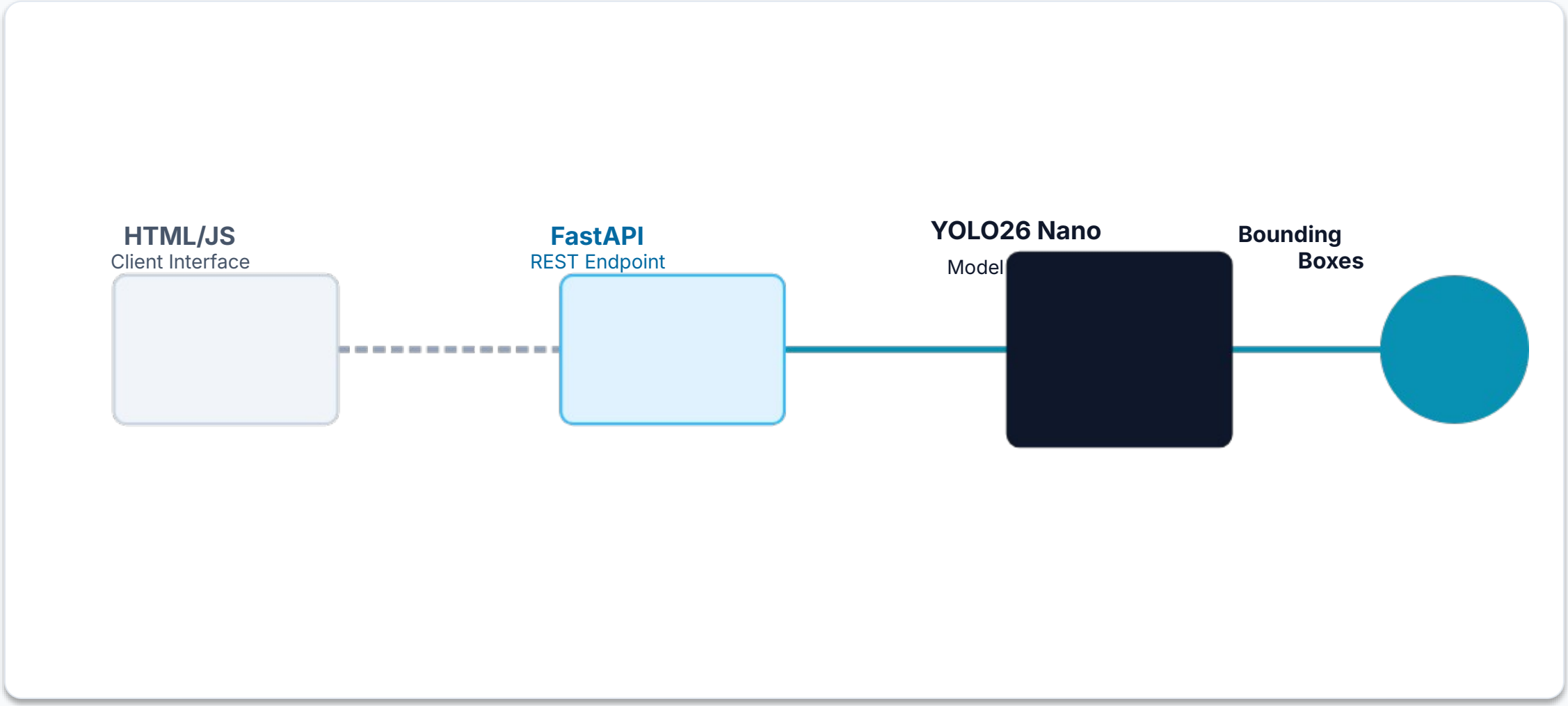


Ultralytics
YOLO26 Nano



Web Frontend

System Architecture



Inference Workflow



1. Capture

Receive raw waste image.



2. Preprocess

Resize and normalize tensors.



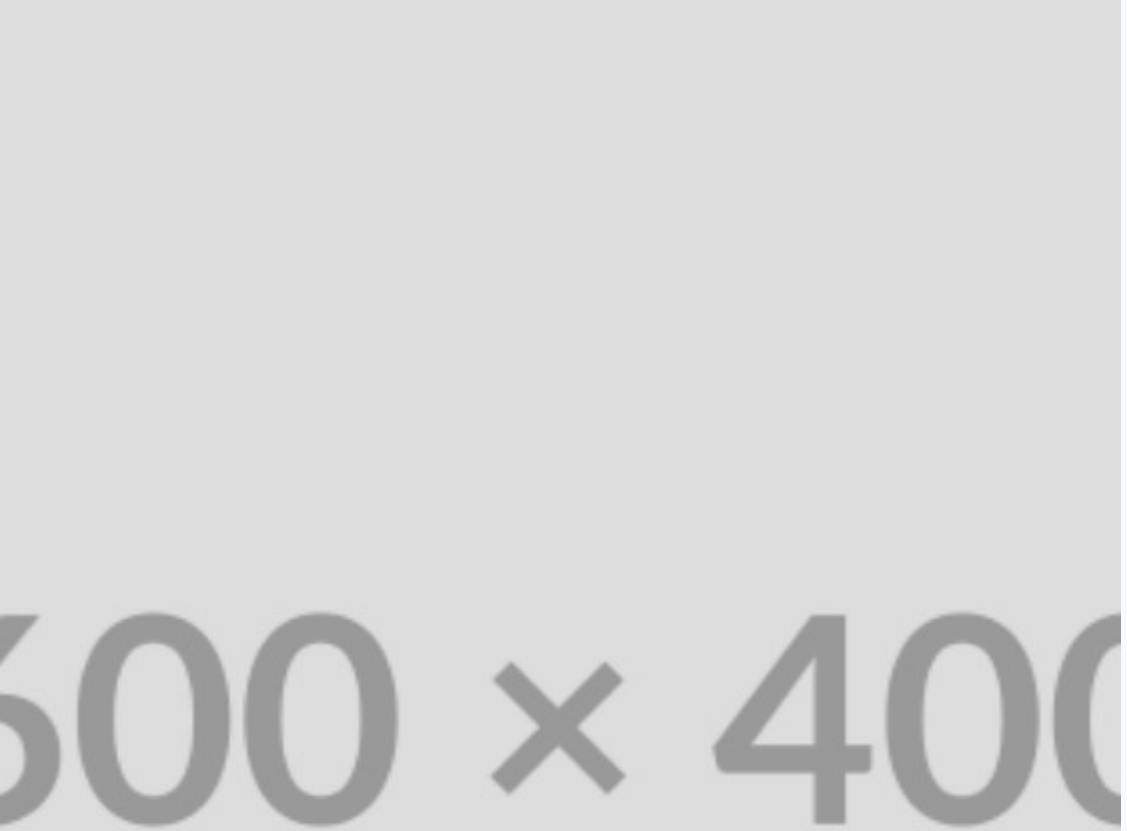
3. Predict

YOLO26 forward pass.



4. Render

Draw boxes on frontend UI.



Detection Algorithm

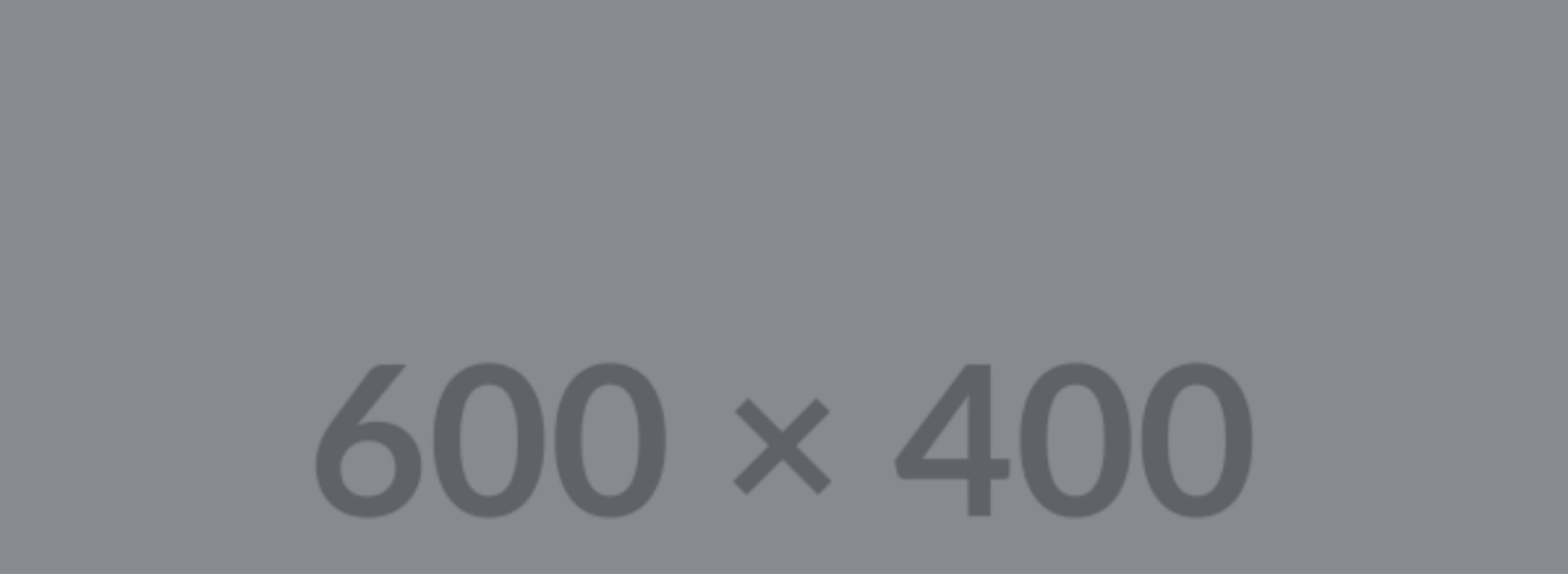
Utilizing Ultralytics YOLO26 Nano architecture for single-stage object detection, optimized for speed and low-compute environments.

- **One-Stage Architecture:** Simultaneous bounding box & class prediction.
- **Nano Variant:** Reduced parameters for real-time edge inference.
- **Augmentation:** Mosaic and mixup training for robustness.

MODEL PERFORMANCE

Held-Out Test Metrics

75.8%	61.6%	69.1%	48.8%
Precision (75.85%)	Recall (61.67%)	mAP@50 (69.14%)	mAP@50-95 (48.87%)



Application Interface

HTML/JS frontend consuming FastAPI inference endpoints to visualize bounding boxes and confidence scores in real-time.

Validation & Limitations

Testing Success

Passed exactly **10 automated application tests**.

API routing, payload validation, and core inference functions verified.

Real-World Limitations

- **Occlusion: Overlapping waste causes missed detections.**
- **Lighting Sensitivity: Shadows misclassify materials.**
- **Class Confusion: Transparent plastic vs. glass errors.**

Technical Q&A

1. Why choose YOLO26 Nano?

Optimizes inference speed for real-time edge deployment constraints.

2. Explain mAP@50?

Mean Average Precision where Intersection over Union (IoU) > 0.50.

3. How does FastAPI improve this architecture?

Provides async handling and high-throughput REST endpoints.

4. Precision vs. Recall trade-off here?

High precision ensures sorting purity; low recall means some recyclables are missed.

5. Mitigation for real-world occlusion?

Requires physical vibratory separation on the belt before camera capture.

Scalable Vision for Sustainability.

Demonstrating the viability of edge-deployed computer vision in solving physical-world ecological challenges.

Local Deployment V1.0

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Image Sources



https://images.unsplash.com/photo-1532996122724-e3c354a0b15b?crop=entropy&cs=tinysrgb&fit=max&fm=jpg&ixid=M3w1ODg1Mjh8MHwxfHNlYXJjaHwyfHxVcmJhbiUyMHdhc3RlJTIwbWFuYWdlbWVudCUyMHBvbGx1dGlvbiUyMHByb2JsZW0lMjBhYnN0cmFjdCUyMHBob3RvfGVufDB8fHx8MTcxOTQwNDQ5MHww&ixlib=rb-4.0.3&q=80&w=1080

Source: unsplash.com



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